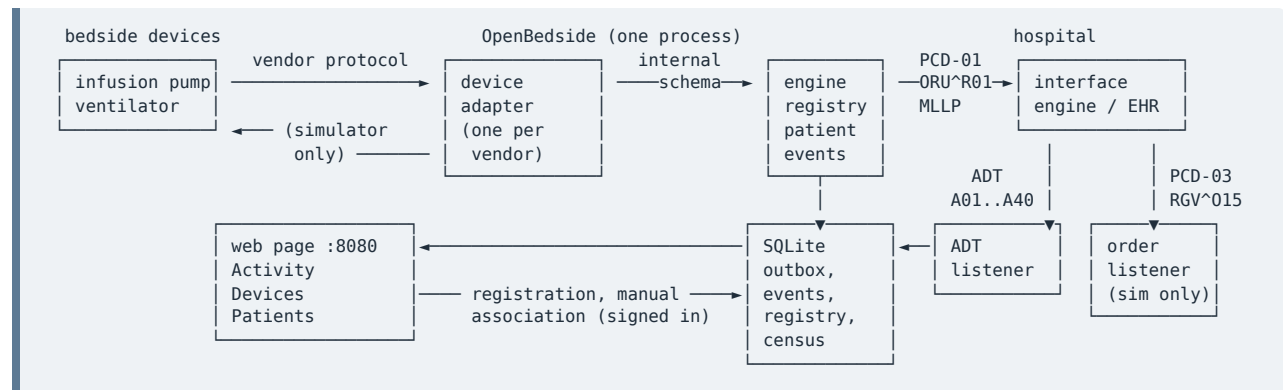


Architecture



The one rule

Vendor protocols stop at the adapter. Everything to the right of the adapter reads only `model.py`. That is what makes the rest identical for every manufacturer, and it is what lets any device be connected by writing one class instead of a gateway.

Pieces

Module	Job	Knows about
<code>model.py</code>	internal schema: Device (MDS) → Module (VMD) → Channel → Source	nothing else
<code>adapters/</code>	talk to devices, publish Device snapshots; <code>StreamAdapter</code> does the connect, read, reconnect loop	one vendor's protocol
<code>validate.py</code>	check snapshots before a hospital sees them (<code>openbedside check</code>)	the schema, the HL7 builder
<code>scaffold.py</code>	write a starter adapter folder (<code>openbedside new-adapter</code>)	nothing
<code>examples/</code>	the fictional ACME pump and its adapter, a complete worked integration	its own invented protocol
<code>core/engine.py</code>	check registration, keep state, derive events, decide what to send	the schema, the outbox, the registry
<code>core/association.py</code>	decide each device's patient, and say how	the registry and census
<code>store/registry.py</code>	registered devices, ADT census, manual associations, admin password	SQLite

Module	Job	Knows about
<code>ehr/adt.py</code>	accept ADT, keep the census (identifiers and location only)	HL7
<code>store/outbox.py</code>	durable queue and event log	SQLite
<code>hl7/</code>	build and parse HL7 v2, MLLP framing, ACKs	HL7, MDC codes
<code>ehr/sender.py</code>	deliver the queue, in order, on application ACK	MLLP
<code>orders/piv.py</code>	accept PCD-03, validate, route (simulator only)	HL7, adapters
<code>ui/server.py</code>	Activity, Devices and Patients pages; sign-in; JSON API	the state snapshot, the registry

Server and interface in one

Earlier commercial gateways split the device server from the interface engine and ran a full SQL database server, for reasons of history as much as design. Here they are one process, and the database is a single SQLite file next to it. A hospital, or a developer on a laptop, installs one thing.

Delivery guarantees

- Every outbound message is written to disk before it is sent.
- A message is delivered only when the receiver returns an application ACK (AA or CA) that names its control id.
- Order is kept per destination. If the oldest message cannot be delivered, nothing behind it goes out.
- An explicit reject (AR) is set aside for a person to review and the queue moves on.
- Everything else backs off and retries, up to five minutes between tries.

Events: what is observed and what is inferred

IHE's infusion event profile (IPEC) requires three events: delivery start, stop and complete. It has **no** rate change, **no** secondary-to-primary switchover and **no** transition to KVO, which are the states a clinician needs to read a pump record. The engine infers all of these from successive snapshots and labels every one `derived`, so nothing downstream mistakes an inference for a device report.

Cloud later, without redesign

The outbox is addressed by destination. A second sender can drain the same queue to an HTTPS endpoint, a message broker or a cloud service, alongside or instead of the EHR. Nothing in the adapters or the engine changes. Whether to do that is a market question this project will

answer when hospitals do.

Traps the design handles on purpose

- **Rate is not dose.** A dose rate is sent only with the concentration, and weight when weight-based, that produced it. Missing basis means the dose rate is withheld.
- **KVO is still infusing**, at a rate unrelated to the order. It is its own status.
- **A secondary pauses the primary** on a shared line. Only the active source's rate is the channel's actual rate.
- **Multi-module pumps are one device.** Two modules are two VMDs under one MDS.
- **Patient identity** is never guessed. It comes from sources the registration allows (the device, a person, the ADT census for the device's bed), and a disagreement means `Unknown`. See `core/association.py`.
- **Registration before transmission.** An unregistered device is held and shown on the Devices page, not sent. Identity is vendor plus device id, never an IP address.
- **Clocks drift.** The model keeps both the gateway's observation time and the device's own clock so skew can be seen.